

FLOWRA — DigitalOcean Droplet Deployment Guide

One-shot, copy-paste guide to deploy FLOWRA on a fresh Ubuntu 22.04 / 24.04 droplet with HTTPS at <https://insights.flowralive.in>. Backend stays at FastAPI on port 8001 (proxied by nginx); frontend is served as a static React build by the same nginx. MongoDB Atlas is already live, so the droplet only runs API + static files.

Time to first request: ~30 minutes including DNS propagation.

0. Before you start (5 min)

You'll need: - DO droplet with public IP (you already have one). - Root or sudo SSH access. - A laptop with a terminal. - Your GoDaddy / Cloudflare / wherever-you-registered-flowralive.in DNS panel open in a browser tab.

0a. Point the DNS record

In your DNS panel, add this A record:

Type	Name	Value	TTL
A	insights	<your droplet public IP>	300

Save. DNS propagates in 1-10 min globally. Verify from your laptop:

```
dig +short insights.flowralive.in
# → should print the droplet IP
```

1. SSH into the droplet

```
ssh root@<your droplet ip>
```

2. Install system dependencies (one-time, ~3 min)

```
apt update && apt upgrade -y

# Core utilities
apt install -y git curl ufw nginx certbot python3-certbot-nginx \
    build-essential

# Python 3.11 (the version your code targets)
apt install -y python3.11 python3.11-venv python3.11-dev python3-pip

# Node 20 (for the React build)
curl -fsSL https://deb.nodesource.com/setup_20.x | bash -
apt install -y nodejs
npm install -g yarn

# Firewall (only HTTP/HTTPS + SSH open)
ufw allow OpenSSH
ufw allow 'Nginx Full'
ufw --force enable
```

3. Clone the app

Use DigitalOcean console's GitHub clone, or:

```
mkdir -p /opt && cd /opt
git clone https://github.com/<your-username>/<your-repo>.git flowra
cd flowra
```

If you've already cloned into a different folder, just `cd` there. The rest of this guide assumes `/opt/flowra` — adjust paths if different.

4. Configure environment variables

4a. Backend `.env`

```
cat > /opt/flowra/backend/.env <<'EOF'
MONGO_URL="mongodb+srv://jodidarindia_db_user:0TTtFS0rJLz3DdTE@flowra-cluster.cxt8yw1.mongodb.net/?retryWrites=true&appName=flowra"
DB_NAME="Flowra-Insights"
CORS_ORIGINS="https://insights.flowralive.in"
EMERGENT_LLM_KEY=sk-emergent-aC1Cb333033447c9d9
RESEND_API_KEY=re_YwNsMM4h_GSgBpnEQpyBRV1ieR6oeyGFU
SENDER_EMAIL=support@flowralive.in
JWT_SECRET=b49043023b7d21ed66c0d0109ca760872b4f71acd0774edbcef786f72633c55e
ADMIN_USERNAME=admin
ADMIN_PASSWORD=admin123
SUPER_ADMIN_USERNAME=superadmin
SUPER_ADMIN_PASSWORD=superadmin123
ENCRYPTION_KEY=0XxGtFVpZUOUVYtYTt6rI7jsYWsPh2Wc6pg0b3n_PcE=
RECAPTCHA_SECRET_KEY=6LeoJLMsAAAAAJK_oy4L6BE8C3fimRDTNrZ677u7
EOF
chmod 600 /opt/flowra/backend/.env
```

⚠ Change ADMIN_PASSWORD and SUPER_ADMIN_PASSWORD before going live! Use a 20+ character random string. The current values are demo defaults.

4b. Frontend `.env`

```
cat > /opt/flowra/frontend/.env <<'EOF'
REACT_APP_BACKEND_URL=https://insights.flowralive.in
REACT_APP_RECAPTCHA_SITE_KEY=6LeoJLMsAAAAANrpgqaHnjBtYTY4ob1dJniDyA1E
EOF
```

5. Backend — Python venv + dependencies

```
cd /opt/flowra/backend
python3.11 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
# Sanity check - should print 51 collections and 7,150+ docs
python -c "
import asyncio, sys
from dotenv import load_dotenv
load_dotenv('.env')
from db import db
async def m():
    cols = await db.list_collection_names()
    print(len(cols), 'collections')
asyncio.run(m())
"
deactivate
```

6. Frontend — build the React static files

```
cd /opt/flowra/frontend
yarn install
yarn build
# Produces /opt/flowra/frontend/build/ - this is what nginx will serve.
```

7. systemd unit for the backend

```
cat > /etc/systemd/system/flowra-backend.service <<'EOF'
[Unit]
Description=FLOWRA FastAPI Backend
After=network.target

[Service]
Type=simple
User=root
WorkingDirectory=/opt/flowra/backend
EnvironmentFile=/opt/flowra/backend/.env
ExecStart=/opt/flowra/backend/.venv/bin/uvicorn server:app --host 127.0.0.1 --port 8001 --workers 2
Restart=always
RestartSec=5
StandardOutput=append:/var/log/flowra-backend.log
StandardError=append:/var/log/flowra-backend.log

[Install]
WantedBy=multi-user.target
EOF

systemctl daemon-reload
systemctl enable flowra-backend
systemctl start flowra-backend

# Verify
sleep 3 && systemctl status flowra-backend --no-pager | head -15
curl -s http://127.0.0.1:8001/api/health
# → {"ok": true, "db": "connected", "service": "flowra-backend"}
```

8. nginx config — serves frontend + proxies API

```

cat > /etc/nginx/sites-available/flowra <<'EOF'
server {
    listen 80;
    listen [::]:80;
    server_name insights.flowralive.in;

    # Increase upload limit for Tally syncs (some manifests can be large)
    client_max_body_size 50M;

    # ——— Static frontend ———
    root /opt/flowra/frontend/build;
    index index.html;

    # ——— API + WebSocket proxy ———
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_set_header Host          $host;
        proxy_set_header X-Real-IP      $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 300s;
        proxy_send_timeout 300s;
        proxy_buffering off;
        proxy_request_buffering off;
    }
    location /ws/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_http_version 1.1;
        proxy_set_header Upgrade      $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host          $host;
        proxy_read_timeout 86400s;
    }

    # ——— React Router fallback ———
    location / {
        try_files $uri /index.html;
    }

    # Static asset caching
    location ~* \.(?:js|css|woff2?|ttf|png|jpg|jpeg|gif|svg|ico)$ {
        expires 30d;
        add_header Cache-Control "public, immutable";
        try_files $uri =404;
    }

    # Plain-text files (PDFs, .exe etc.) – no caching so updates show immediately
    location ~* \.(?:pdf|exe|zip)$ {
        expires 0;
        try_files $uri =404;
    }
}
EOF

```

```
ln -sf /etc/nginx/sites-available/flowra /etc/nginx/sites-enabled/flowra
rm -f /etc/nginx/sites-enabled/default
nginx -t && systemctl restart nginx
```

9. HTTPS — free SSL via Let's Encrypt

```
certbot --nginx -d insights.flowralive.in \
    --non-interactive --agree-tos -m support@flowralive.in --redirect
```

Certbot rewrites the nginx config to add the HTTPS server block and a 80→443 redirect. Auto-renewal is set up automatically (twice-daily systemd timer).

Verify:

```
curl -s https://insights.flowralive.in/api/health
# → {"ok": true, "db": "connected", "service": "flowra-backend"}
```

10. Smoke test

Open `https://insights.flowralive.in/` in a browser.

Log in with: - `admin / admin123` (replace with your new password from step 4a) - or `demo@flowralive.in / demo2026` (the pre-seeded demo tenant)

If you see the dashboard with ₹39.82L total sales → you're live. 🎉

11. Tally Sync Agent — what URL to feed it?

After deploy, when distributing the agent EXE:

- The agent's pre-filled URL (`DEFAULT_BACKEND_URL` constant) is already `https://insights.flowralive.in` as of v9.8.19.
- End-users do NOT need to type a URL — Settings tab pre-fills it.
- Existing v9.8.18 agents already in the field: tell users to go to Settings → Advanced → FLOWRA Server URL and replace the old preview URL with `https://insights.flowralive.in`, then click Save & Start Sync.

Just upload the new `FlowraTallyAgent_v9.8.19.exe` to `/opt/flowra/frontend/build/FlowraTallyAgent.exe` (nginx serves it from the build folder), and customers downloading from the Setup page will get the correctly-configured binary.

12. Day-2 ops cheat sheet

Need to do	Command
Backend logs	<code>journalctl -u flowra-backend -f</code> or <code>tail -f /var/log/flowra-backend.log</code>
Backend restart	<code>systemctl restart flowra-backend</code>
Nginx restart	<code>systemctl restart nginx</code>
Re-pull from GitHub	<code>cd /opt/flowra && git pull && cd backend && source .venv/bin/activate && pip install -r requirements.txt && deactivate && cd ../frontend && yarn install && yarn build && systemctl restart flowra-backend && systemctl reload nginx</code>
Renew certs (auto)	<code>certbot renew --dry-run</code> — should say "Congratulations"
Check Atlas connection	<code>curl https://insights.flowralive.in/api/health</code>

13. Common pitfalls

Symptom	Cause	Fix
<code>/api/health</code> returns 502	Backend down	<code>systemctl status flowra-backend</code> → check logs
<code>db: disconnected</code> in health	Atlas IP allowlist	Add droplet IP (or 0.0.0.0/0) to Atlas → Network Access
Login works but dashboard empty	Frontend rebuilt with wrong URL	Rebuild frontend after fixing <code>frontend/.env</code>
Agent says "Cannot reach FLOWRA"	DNS not propagated yet	<code>dig insights.flowralive.in</code> — wait 5-10 min
reCAPTCHA fails on login	Site key restricted to localhost in Google console	Add <code>insights.flowralive.in</code> to allowed domains in https://www.google.com/recaptcha/admin
EXE 404	Build folder doesn't contain it	<code>cp /opt/flowra/frontend/public/FlowraTallyAgent.exe /opt/flowra/frontend/build/</code> then <code>systemctl reload nginx</code>

14. What's left to harden post-launch

- Database snapshots: Atlas → Backup → enable daily snapshot retention (free tier = 2/day).

- Off-server log rotation: `apt install -y logrotate` (already installed); add a `/etc/logrotate.d/flowra-backend` if logs grow large.
- Password rotation: change `ADMIN_PASSWORD` and `SUPER_ADMIN_PASSWORD` to 20-char random strings.
- WhatsApp + GST integrations: still on the roadmap — don't block production launch waiting for these.
- Monitor uptime: free option = `https://uptimerobot.com` → add `https://insights.flowralive.in/api/health` and get an email if it ever goes down.

Last updated: May 2026 · FLOWRA v9.8.19 / Agent v9.8.19